

Sergio Córdoba, MSc.

Data | AI | Analytics
From data to business assets

PORTFOLIO:

<https://sergiocy.github.io/>

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About me...

With mathematical background and 10 years of expertise around data, i have an strong analytical (and business) thinking and technical capabilities, that i apply to get production-ready AI or ML systems in high-scale environments and data-driven companies, where there are an economic impact.

I have designed and deployed ML pipelines and AI systems in production, including segmentations, optimization processes, recommendation systems and RAG-based internal knowledge systems, owning the full lifecycle, playing a cross-functional role and engaging with stakeholders.

Languages...

Spanish; native
English; conversational
French; basic

Academic background

Postdegree in "Advanced methods in applied statistic" (UNED, Spain, 2017)

MSc. in "Mathematical modeling and research, statistics and computing" (University of Zaragoza, Spain, 2015)

Postdegree in "Microsystems and smart instrumentation" (University of Zaragoza, Spain, 2013)

MSc. in Physics (spanish 5-years-degree. University of Zaragoza, Spain, 2013)

Workshops and courses

#tensorflow #datarobot
#IBMWatson #neuralnetworks
#hadoop #spark #agile #linux
#simulation
#computationalscience
#softwarearchitecture

Experience

*** Senior/lead data scientist at Vueling Airlines / KnowMadMood, data&AI dept, 2018-2025, Zaragoza (Spain)

I have developed data products for distinct business areas (sales, customer, digital)

- interaction with stakeholders to understand the use-case and business goal
- reviewing state-of-the-art around the use-case to define the **model approach**.
- defining tasks, timelines and deliveries (**jira, scrum, crisp-dm, kanban**)
- performing business presentations; descriptives, results, business-cases, scenarios,
- Reporting, ad-hoc analysis (**powerpoint, matplotlib, seaborn, plotly, excel, jupyter**)
- Project ownership, tracking and monitoring the model performance and linking with stakeholders to analyze and monetize the output.
- Hybrid and cross-functional expertise along data and cloud. Applied ML-engineering. ntegration with IT, data-engineering and MLOps teams.

I design, build and industrialize end-to-end ML or AI pipelines (training, predict),

- data extraction and preparation, feature engineering, cleaning, encoding, preprocessing, sampling. **ETL, SQL**, core in **AWS-Redshift + AWS-S3. Big data**.
- Exploratory Data Analysis (**EDA**); features descriptives, business insights. Univariate, multivariate, correlations, interactions. **Python, pandas, numpy, matplotlib**
- Modeling and evaluation; sampling, stratification, feature selection, performance metrics (**F1, MAE, ROC**, etc). **Jupyter, sklearn, pandas, numpy, scipy**.
- Deployment; as batch-process (**AWS-ECR, docker, airflow**) and API-REST via (**AWS-APIGateway, postman, flask, JSON, AWS-Lambda**) .. Also, **AzureDevOps**.
- Monitoring and maintenance; logs, alerts, mailing, business KPIs (conversion-rate, revenue, RASK, yield, churn-rate). **python, dash, grafana, datadog, kibana, tableau**

Along the time i have been part of a team's growth and have faced different challenges technological and functional,

- * 2025 – **genAI, RAG, LLMs and LLMOps (Azure OpenAI, Azure DevOps)**
RAG to load company documentation (~7000 pages)
- * 2023 – **personalization tech-lead**, supervision and mentoring. Tasks planning and reporting.
Set of projects: **new customers datamart + binary classifier + clustering + segmentation + life-time value (xgboost, lightgbm, kmeans, kprototypes, multiclass, datadog, grafana)**
IT WAS REFERENCE OF CUSTOMERS PERSONAL INFO AND HISTORY
- (PoCs) integration of web activity and marketing automation (**GCP, GA4, UA, GoogleAnalytics, CloudStorage, CloudFunctions, BigQuery**)
- * 2021 – **propensity to purchase** from web-activity and **drivers of purchase repetition**
- product-recommender; integration DS + IT-Mobile team, **AWS-Sagemaker**
- * 2020 – (bag) **pricing optimization (regression, demand, xgboost, ONNX, datarobot, A-B testing, elasticity)**
- Work with **distributed teams and advisors (Colombia, USA)**.
FIRST MODEL DEPLOYED IN PRO IN (3rd party) **PRICING PLATFORM**
AN INCREMENT OF 10% OF REVENUE (FROM BAGS) WAS ESTIMATED
IT WORKED IN PRODUCTIVE WEB PROVIDING LUGGAGE PRICES
- datamarts maintenance, moniktoring, **python. dash**
- * 2018 – **Starting team** (i was first person at data dept in Zaragoza Office, distributed team, remote work); **Heterogeneous sources, systems and processes**.
- involved in **project standarizacion and MLOps initiatives** (project benchmarks, MLflow deployment, data-engineering flows maintenance)
- Working over **cloud-server (backend); ssh, mobaxTerm, winscp, scp, linux, bash, AWS-EMR, AWS-EC2**
- **Anomaly-detection** project; **R-language**, deployed over **AWS-EMR**. Scheduled and executed with **bash script and cron**. Later, orchestration with **MLflow demand forecast pipelines. R-language, python, hive, pyspark, sparkR, spark, sparkSQL, hdfs, parquet**

*** PhD courses at UNED, 2018-2021, Madrid (Spain)

"Intelligent Systems" program. Around algebraic treatments of word-embeddings. (**python, R, nltk, BERT, Word2Vec, GloVe, text-similarity, LLM, research, wrtting**)

*** Data scientist at NTT Data (everis), 2016-2018, Zaragoza (Spain)

Java, R-language, ontologies, semantic searcher, NLP, web-scraping, graphs data bases, SolR, elasticSearch, **graphDB, OWL, RDF-storage, sparQL**, pentaho, kibana

*** BI consultant at Banco Sabadell (Innova) 2016, Barcelona (Spain)

*** Graduate research fellow, UNIZAR-IUMA, 2015, Zaragoza (Spain)